The Chain Rule

Basic chain rule applications

1 Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

b $f(x) = \sin(3x^2)$

c $f(x) = e^{5x}$

d $f(x) = \ln(2x + 7)$

2 Differentiate $f(x) = \sqrt{x^2 + 9}$.

The chain rule with trig and exponential functions

3 Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

4 Differentiate $f(x) = e^{\sin x}$.

5 Differentiate $f(x) = \tan(e^x)$.

Nested (multiple) chain rule

6 Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.

7 Differentiate $f(x) = (\ln(3x))^4$.

Chain rule combined with product/quotient rules

8 Differentiate $f(x) = x^2 e^{3x}$.

9 Differentiate $f(x) = \frac{\sin(2x)}{x}$.

Chain rule with tables of values

10 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

Chain rule with inverse trig and logarithms

11 Differentiate $f(x) = \sin^{-1}(2x)$.

12 Differentiate $f(x) = \ln(\sin x)$.

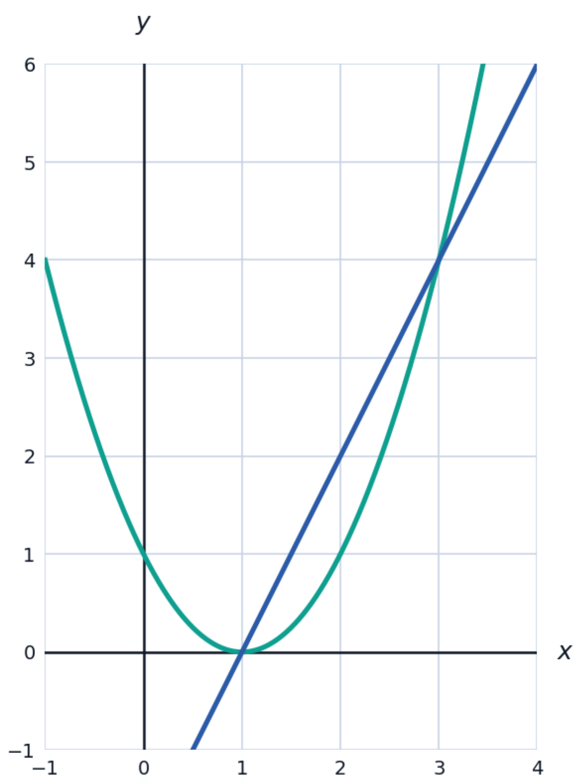
13 Differentiate $f(x) = \tan^{-1}(x^2)$.

Finding tangent lines using the chain rule

14 Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

Visualizing a composite function and its derivative

15 The graph shows $f(x) = (x - 1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.



a Use the chain rule to find $f'(x)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown.

The chain rule in applied rate problems

16 The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.

Chain rule with a table of composed values

17 Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.

18 Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.